

ALGONQUIN PARK EAST GATE, ON, Canada

WMO: 715810

Lat: 45.536N

Lon: 78.262W

Elev: 397

StdP: 96.65

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.6	(c) -27.2	(d) -33.9	(e) 0.2	(f) -30.3	(g) -30.5	(h) 0.2	(i) -27.0	(j) 4.1	(k) -5.3	(l) 3.5	(m) -6.3	(n) 0.1	(o) 0	(p) 0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	29.1	19.5	27.5	18.7	26.0	17.8	21.5	26.3	20.6	25.0	19.7	23.7	2.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	15.5	23.4	19.1	14.6	22.6	18.1	13.7	21.8	64.5	26.2	61.2	24.9	58.0	23.6	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3.5	3.3	2.9	-36.2	32.0	2.6	1.5	-38.0	33.1	-39.5	33.9	-41.0	34.7	-42.8	35.8
			-36.3	23.4	2.6	1.1	-38.1	24.2	-39.6	24.9	-41.1	25.5	-42.9	26.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-11.3	-10.6	-4.9	3.1	10.9	15.7	18.2	17.0	12.8	6.4	-0.6	-7.7
	DBStd	11.79	7.87	6.69	6.97	4.65	4.77	3.63	3.05	3.04	4.25	4.54	5.32	6.84
	HDD10.0	2981	662	577	464	212	47	2	0	0	19	132	319	547
	HDD18.3	5261	920	810	722	457	237	96	39	62	172	371	569	806
	CDD10.0	848	0	0	1	4	73	173	256	217	103	21	0	0
	CDD18.3	87	0	0	0	0	6	17	37	21	7	0	0	0
	CDH23.3	1111	0	0	1	6	108	221	446	247	78	5	0	0
	CDH26.7	231	0	0	0	1	26	44	106	44	11	0	0	0
(14) Wind	WSAvg	1.0	1.0	1.1	1.2	1.2	1.1	1.0	0.9	0.8	0.8	1.0	1.0	1.0
(15) Precipitation	PrecAvg	953	80	59	64	66	78	81	85	77	94	91	90	80
	PrecMax	1100	133	106	157	127	123	131	136	130	164	145	162	126
	PrecMin	822	43	20	12	20	33	22	29	22	32	15	54	30
	PrecStd	79	28	24	33	28	23	29	31	23	30	36	28	22
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	7.1	15.5	23.5	29.7	29.9	31.8	30.5	28.2	23.8	15.5	9.3
		MCWB	8.4	5.1	10.2	13.5	18.6	20.4	20.4	20.7	20.3	16.7	11.4	8.2
	2%	DB	4.0	3.5	10.4	18.6	26.3	28.0	29.4	27.6	25.6	20.0	11.7	5.3
		MCWB	3.0	2.3	5.6	9.8	16.4	18.9	19.7	19.0	19.0	14.4	8.9	4.5
	5%	DB	1.4	1.1	7.5	15.2	23.6	26.0	27.5	26.0	23.5	16.9	9.4	2.5
		MCWB	0.8	-0.1	3.6	8.6	14.1	17.6	18.8	18.2	17.4	13.4	7.4	1.7
	10%	DB	-0.1	-0.3	4.7	11.9	20.9	23.9	25.8	24.4	21.3	14.4	7.1	0.7
		MCWB	-0.8	-1.2	1.6	5.9	13.4	16.6	17.9	17.5	16.7	11.6	5.4	0.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.3	5.9	10.9	14.9	20.4	22.2	23.0	22.2	21.5	18.4	12.5	8.8
		MCDB	8.8	6.5	13.1	20.2	27.4	27.2	28.2	26.7	25.6	21.0	14.5	9.3
	2%	WB	3.2	2.3	6.6	12.1	18.3	20.8	21.6	21.0	20.3	15.9	9.5	4.6
		MCDB	3.9	3.7	9.5	16.9	23.5	25.7	26.4	25.1	24.2	19.1	11.1	5.2
	5%	WB	0.9	0.3	3.9	8.9	16.7	19.5	20.5	19.9	18.9	13.8	7.5	1.8
		MCDB	1.3	0.9	6.4	13.5	20.8	23.4	24.9	23.5	21.8	16.4	9.1	2.4
	10%	WB	-0.8	-1.3	2.1	7.0	14.6	18.1	19.6	18.8	17.4	11.7	5.5	0.2
		MCDB	-0.1	-0.4	4.3	10.9	18.7	21.7	23.8	22.6	20.4	13.9	6.9	0.6
(35) Mean Daily Temperature Range	MDBR		11.9	12.8	13.5	12.4	14.3	12.8	13.2	12.8	13.2	10.1	9.1	9.3
	5% DB	MCDBR	9.6	11.6	15.4	19.7	19.2	16.2	15.8	15.3	15.7	15.4	12.9	8.1
		MCWBR	9.1	10.1	10.6	11.6	9.7	8.0	7.5	7.9	9.4	10.2	10.1	7.5
	5% WB	MCDBR	9.4	10.4	13.5	16.7	14.3	12.9	12.8	12.0	12.7	13.2	11.7	7.9
		MCWBR	9.1	9.6	10.1	11.1	8.2	7.3	7.0	7.1	8.3	9.5	9.9	7.4
(40) Clear-Sky Solar Irradiance	taub		0.269	0.279	0.300	0.337	0.352	0.390	0.391	0.380	0.357	0.334	0.297	0.267
	taud		2.366	2.330	2.430	2.431	2.434	2.342	2.351	2.380	2.436	2.491	2.531	2.419
	Ebn at Noon		847	907	937	924	916	878	871	868	861	834	815	813
	Edh at Noon		73	93	98	107	111	122	120	112	97	80	64	63
(44) All-Sky Solar Radiation	RadAvg		1.30	2.21	3.63	4.51	5.19	5.68	5.70	4.94	3.85	2.27	1.32	0.96
	RadStd		0.17	0.27	0.36	0.38	0.48	0.46	0.43	0.29	0.37	0.24	0.16	0.12

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	-1.22	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page